

Experienced data scientist at Vanguard with 7+ years in data science, data engineering, and software engineering, adept at turning complex data into actionable insights & business gains, proven through significant contributions at Vanguard, BAIR Labs and Walmart.

EDUCATION

Master’s in Information and Data Science GPA - 4.0/4.0, University of California, Berkeley	Graduated: Dec 2024
Master of Science, Computer Engineering, GPA - 3.42/4.0, San Jose State University	Graduated: Dec 2018
Bachelor of Engineering, Electrical and Electronics, GPA - 3.90/4.0, Anna University	Graduated: June 2014

SKILLS

Languages	Python, R (R Studio), Scala, Golang
Database	SQL, SAS, PostgresSQL, MySQL, Redis, Mongo, Neo4j
Data Science & Analytics	RAG, Time-series Analysis, Hypothesis Testing, Statistics, Probability, A/B Testing, PCA, Linear Regression, Logistic Regression, Decision Tree, Random Forest, Clustering, Naive Bayes, Predictive modeling, XGBoost, GBM, KNN, SVM, Neural Networks(DNN, CNN, SSD mobilenet, GRU, RNN, LSTM), NLP, Ensemble, Large Language Models(LLM), Transformers, MLP, GBT, Cross-Validation, Text Classification, GPT-4, Multi-modal AI, BERT, Roberta, Llama-2, Graph Neural Networks (GNN),AWS (Fraud Detector, Neptune DB), sagemaker, Cohere, Claude, Qdrant, PyTorch
Frameworks/Web Dev	LangChain, TensorFlow, Hugging Face, PyTorch, Keras, Django, Flask, React JS, HTML, CSS
Data Analysis	Pandas, NumPy, Scikit-Learn,Scipy, librosa, Audio Feature analysis, Excel, Jupyter Notebook
Big Data	Apache Spark, Kafka, Akka stream, Elastic Search, Hadoop, MapReduce, ETL
Data Visualization	Matplotlib, Seaborn, Plotly, D3.js, Altair, Tableau, Power BI, Grafana, Splunk, Amcharts, ggplot(R), Looker
Cloud	AWS, Google Cloud Platform (GCP), Databricks, AWS Sagemaker
OS/Scripting	Linux, Bash, Windows
Containers/Orchestration	Docker, Kubernetes, CI/CD
Version Control	Git

WORK EXPERIENCE

Data Scientist , - Vanguard, Malvern, PA	Jul 2024 - present
- Designed Narwhal, a graph-based model, to detect fraud rings and enhance capture rates through transaction and behavior analysis. - Designed AWS Fraud Detector pipeline across Vanguard journeys to detect new account fraud and minimize good user decline below 5%. - Designed the Open Smart Learning model on LexisNexis, benchmarking against AWS FD to improve fraud detection accuracy and efficiency. - Led model refreshes, refinement of feature engineering, production evaluation, and deployment. - Conducted root cause analysis during large-scale fraud attacks, inspecting false negatives and false positives to refine models. - Drove \$100K+ in fraud cost savings and a 15% reduction in fraud losses through proactive model improvements. - Developed a GenAI-powered Risk Scan Model for instant risk and compliance queries, streamlining stakeholder access to regulations. - Integrated the Risk Scan Model with Vanguard’s knowledge base, boosting accuracy by 30% and cutting manual effort by 50%. - Mentoring new talents in fraud analytics.	
Machine Learning Researcher , Berkeley AI Research (BAIR) Labs (Advisor: Irene Chen)	Dec 2023 – June 2024
- Led research on climate change’s health impacts, uncovering environmental-health links to drive insights and policy actions. - Built predictive models using graph neural networks to correlate climate anomalies with health risks, boosting prediction accuracy. - Analyzed high-dimensional medical data from UCSF and HCUP to identify key health risk factors tied to climate anomalies.	
Software Engineer, Analytics - Walmart eCommerce, Sunnyvale, CA	Jan 2019 - Jan 2023
- Solved large-scale data engineering problems using big data and streaming technologies to enhance the Walmart store ecosystem. - Designed microservices for store infrastructure incident resolution, utilized AIOps to detect and diagnose issues, improving MTTR by 2.5x. - Enhanced efficiency by 42% with an ML predictive system for anomaly detection, scaling support for 20,000 users. - Led product development and deployed models into production, leveraging high-quality data, data-driven design, and stakeholder feedback to deliver robust, scalable solutions. - Developed and executed complex SQL queries to uncover core infrastructure insights and correlate data.	
Devops Fellow Engineer - Aeris Communications,San Jose, CA	Oct 2018 - Dec 2018
- Designed and implemented tools for automating the deployment of IOT applications. - Created automation to handle disk and memory log errors from Nagios and ELK stack.	
Machine Learning Engineering Intern - Konviv Inc, Berkeley, CA	May 2018 - Aug 2018
- Developed an AI-based financial management chatbot through K-Means clustering for customer transaction categorization. - Worked on Bayesian network based recommendation engine to guide the customers for better financial decisions.	
Graduate Deep Learning Researcher - Tower Foundation of SJSU, San Jose, CA	May 2017 - Aug 2018
- Contributed to UAV graffiti removal system development in San Jose, achieving 90% graffiti detection accuracy with SSD Mobilenet. - Employed inception model to differentiate graffiti and non-graffiti images for drone classification.	
Software Engineer - Larsen & Tubro (L&T) Infotech, Chennai, India	July 2014 - June 2016
- Developed COBOL modules and SQL queries, driving an 80,000 increase in the online travel insurance customer base. - Rectified the production defects and played a major role in improving the latency of screen designs.	

DATA SCIENCE PROJECTS

Policy Pulse Project (GDPR Compliance AI Tool) UC Berkeley, CA

Aug 2024 – Dec 2024

- Lead and developed GenAI RAG application to automate GDPR compliance for fintechs, reducing manual review time by 50%.
- Collaborated with SMEs to generate ground truth data and validate model outputs.
- Built a QA system using Cohere Command R+ model achieving 72% accuracy in GDPR violation detection.
- Integrated AWS SageMaker and HuggingFace embeddings, and summarization of privacy policies.
- **Technologies Used:** DeBERTa, RAG, LLM, Cohere, LLaMA2, Mistral AI, Falcon

Enhancing Interdisciplinary Research Using SciBERT-CNN and Topic Modeling UC Berkeley, CA

Jan 2024 – May 2024

- Engineered a novel hybrid SciBERT-CNN model with topic modeling, achieving a 0.71 F1 score on a scientific corpus to enhance research.
- **Under review** at the International Conference on Knowledge Discovery and Data Mining (KDD 2025) - ([Link to paper here](#))
- **Technologies Used:** SciBERT, CNN, RNN, LSTM, BERT, Roberta, Longformer, LLaMA-2/Mistral AI, BERT, T-5

Metastatic Cancer Diagnosis Prediction Women In Data Science Datathon (WIDS), 2024

Jan 2024 – Feb 2024

- Spearheaded a three-member team to promote healthcare equity, by pioneering the analysis of oncology data to predict metastatic cancer.
- Achieved a leading AUC score of 80%, showcasing our model's excellence in early diagnosis.
- **Technologies Used:** Adasyn, SMOTE, Logistic Regression, Catboost, Random Forest, Xgboost, LightGBM, SVM, Optuna, cross-validation, ensemble, Neural Network, GBT

Real-Time Prediction of Flight Departure Delays UC Berkeley, CA

Oct 2023 – Dec 2023

- Led the development of a predictive analytics pipeline, achieving a 52.26% F2 score in forecasting flight delays using historical data.
- Processed over 2 million high-dimensional records, cleaned data, and created time-frequency features to boost the F2-score.
- Executed hyperparameter tuning via grid search, refining model parameters to bolster precision and recall in predicting flight delays.
- Implemented robust blocked time-series split cross-validation to prevent data leakage, and fine-tuned models using grid search.
- Led data integration, enhanced predictive models with real-time insights, improved decision-making and efficiency in the airline industry.
- **Technologies Used:** Spark, Spark RDDs, Spark MLlib, Logistic Regression, Decision Trees, Random Forest, Naive Bayes, MLP, RNN, Ensemble models, GridSearchCV, Databricks

ScoreScope - Educational Data Visualization Platform UC Berkeley, CA

Oct 2023 – Dec 2023

- Developed ScoreScope, a web platform for visualizing educational data including budgets, enrollments, and student performance.
- Utilized U.S. Census Bureau and NAEP data to provide an in-depth view of student achievement in U.S. elementary and secondary schools.
- Enabled educators and policymakers to make data-driven decisions by providing clear, actionable insights.
- Conducted user experience studies with six participants and refined the web interface for an improved user experience based on feedback.
- **Technologies Used:** Python, Altair, Tableau, HTML, CSS, Javascript, Bootstrap, Plotly

Bird Song Classification Using Neural Networks and Machine Learning UC Berkeley, CA

July 2023 – Aug 2023

- Spearheaded the use of BirdCLEF 2023 Kaggle data for bird species classification and biodiversity monitoring.
- Used audio augmentation with machine learning models to improve generalization and extract key features like MFCC and chroma.
- Achieved 95% training and 87% testing accuracy using GRU RNN with evaluation based on F1-score metrics.
- **Technologies Used:** Python, Tensorflow, Logistic Regression, Random Forest, SVM, Xgboost, CNN, LSTM, GRU RNN

Revolutionizing Acmet Gourmet Meal (AGM) Delivery with NoSQL Data Magic UC Berkeley, CA

July 2023 – Aug 2023

- Led a transformative project at AGM, integrating NoSQL databases for innovative meal delivery solutions.
- Utilized graph algorithms (page rank, community detection and closeness centrality) for optimizing BART delivery networks.
- Implemented customer recommendation systems with personalized route suggestions.
- **Technologies Used:** Python, SQL, Neo4j, Graph algorithms, MongoDB, Redis

Big Budgets, Big Returns? - A Film Industry Analysis UC Berkeley, CA

Mar 2023 – Apr 2023

- Analyzed The Movie Database (TMDB) data, building regression models to identify budget-revenue correlations and validate assumptions of large-sample datasets.
- Led a team to develop regression models using covariates like runtime and vote count, identifying the best model with adjusted R^2 , showing potential for a 77.2% revenue increase.
- **Technologies Used:** Python, R, Statistics, Linear Regression, OLS.

LEADERSHIP AND COMMUNITY ENGAGEMENT

Ambassador at Women In Data Science (WIDS)

Jan 2024 - Jan 2025

- Submitted paper on "**Exploring Generative AI for Cyber Threat Intelligence: A Survey with analysis of real-world case studies**" - under review at Artificial Intelligence Review

AWARDS

- Employee Of The Month Award: Awarded the best employee of the month for submitting a paper on "Proactive Incident Management leveraging AIOps techniques" at the internal Walmart conference.

June 2022